

PROJECT PROPOSAL

TOPIC: Hospital Queue Management System Using ANN And ABM

Project Team Members

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1. INTRODUCTION

The Emergency Room environment in hospitals is one of the most complex and sensitive operational areas in healthcare systems because it deals with unpredictable patient arrivals, varying levels of medical urgency, and limited availability of medical resources. In such environments, even small delays can significantly impact patient outcomes, especially in critical cases where immediate attention is required. Traditional queue management approaches, such as first-come-first-served or simple priority-based systems, are not sufficient to handle this level of dynamic variation because they operate on fixed rules and do not adapt to changing real-time conditions. As a result, hospitals often face issues such as overcrowding, inefficient use of medical staff, uneven workload distribution, and increased waiting times for patients. This project addresses these limitations by proposing an intelligent system that combines Artificial Neural Networks and Agent-Based Modeling. The purpose of integrating these two techniques is to create a system that can both understand hospital workload patterns from historical data and simulate realistic hospital behavior under different conditions. The Artificial Neural Network is responsible for learning hidden patterns from emergency room data and predicting congestion levels, while the Agent-Based Model represents real-world hospital operations by simulating how patients arrive, how queues are formed, and how doctors respond to varying workload conditions. This combination allows the system to move beyond static queue rules and behave in a more adaptive and intelligent manner that closely resembles real hospital environments.

2. PROBLEM STATEMENT

The core problem in emergency room management arises from the inability of existing systems to dynamically respond to fluctuating patient inflow and changing workload conditions. In most hospitals, queue handling is managed either manually or through basic priority rules that do not take into account real-time system load, resource availability, or historical patterns of patient arrivals. Because of this limitation, hospitals frequently experience situations where patient queues become excessively long during peak hours while resources remain underutilized during low-demand periods. This imbalance leads to delays in treatment, increased pressure on healthcare staff, and inefficient allocation of medical resources. Additionally, there is no predictive mechanism in place that can anticipate congestion before it actually occurs or simulate

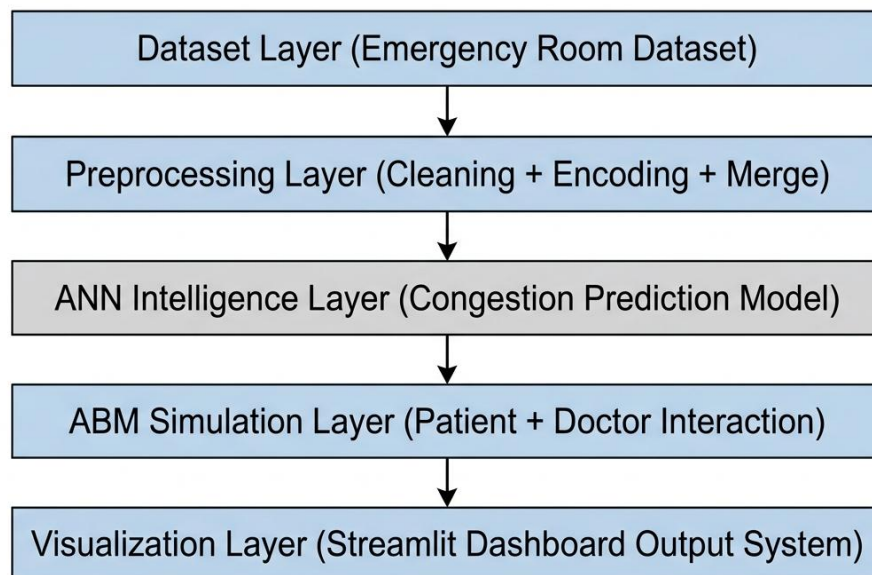
how different load conditions will affect hospital performance. This creates a gap in decision-making support for hospital administrators who need to manage both efficiency and patient care quality simultaneously. Therefore, there is a clear need for a system that can analyze hospital data intelligently, predict congestion levels with reasonable accuracy, and simulate how the hospital environment behaves under different operational conditions.

3. HOW THIS PROJECT SOLVES IT (CORE IDEA)

The system works like this:

WORKFLOW OF SYSTEM

Hospital Queue Management System using ANN and ABM



4. OBJECTIVES

The main objective of this project is to develop an intelligent hospital queue management system that can analyze emergency room data and predict congestion levels using Artificial Neural Networks while also simulating real-world hospital behavior using Agent-Based Modeling. The system aims to improve the understanding of hospital workload patterns and support better decision-making in emergency room operations. It also focuses on reducing patient waiting time by providing a predictive and simulation-based approach to queue management. Another important objective is to classify hospital congestion into different levels such as low, medium, and high so that hospital conditions can be easily interpreted. The system further aims to demonstrate the integration of machine learning and simulation techniques in a healthcare environment for improving operational efficiency.

5. DATASET DESCRIPTION

The dataset used in this project represents real-world emergency room operational data collected from a healthcare monitoring system that records hospital activity over time. It consists of two interconnected components that together describe both structural and behavioral aspects of hospital operations. The first component contains static information about emergency rooms such as hospital name, emergency room identifier, geographical location, and administrative details. This part of the dataset is mainly used for identification and contextual understanding of different hospital locations.

The second component consists of dynamic time-based records that capture actual hospital workload behavior at different time intervals. This includes number of waiting patients, number of examined patients, priority distribution across different urgency levels such as Bianco, Verde, Giallo, and Rosso, and average waiting time recorded at each timestamp. This part of the dataset is crucial because it reflects real fluctuations in hospital congestion and patient flow over time. By combining both static and dynamic information, the dataset provides a complete view of emergency room behavior, making it highly suitable for training the Artificial Neural Network and supporting realistic Agent-Based Simulation.

6. PROPOSED METHODOLOGY (SYSTEM WORKFLOW DESIGN)

PHASE 1 (DATA COLLECTION MODULE)

The system begins by collecting the emergency room dataset which contains both static hospital information and dynamic patient activity records. The static data includes hospital name, emergency room identity, and location details, while the dynamic data includes time-based records such as number of waiting patients, number of examined patients, priority levels, and waiting time. This dataset is loaded into the system in CSV format and serves as the foundation for both prediction and simulation processes.

PHASE 2 (DATA PREPROCESSING MODULE)

In this phase, the raw dataset is converted into a structured format suitable for machine learning. Missing or inconsistent values are removed to ensure data quality. Timestamp values are converted into meaningful time-based features such as hour of the day and workload patterns. Categorical variables such as priority levels are transformed into numerical form so that they can be processed by the model. Irrelevant attributes like hospital contact details and address information are excluded because they do not contribute to congestion prediction. After cleaning and transformation, both datasets are merged using emergency room identification as a common key.

PHASE 3 (FEATURE ENGINEERING MODULE)

After preprocessing, relevant features are extracted to represent hospital workload behavior. These features include number of waiting patients, number of examined patients, priority distribution, and time-based attributes. These selected features form the input space for the Artificial Neural Network and represent the operational state of the emergency room at any given time.

PHASE 4 (ARTIFICIAL NEURAL NETWORK (ANN MODULE))

The Artificial Neural Network is used as the intelligence component of the system. It learns patterns from historical hospital data and identifies the relationship between workload conditions and congestion levels. The input features are passed through multiple hidden layers where the model learns complex relationships between patient load and system congestion. The output of the network is a classification of hospital state into low, medium, or high congestion. This prediction represents the current operational condition of the emergency room and becomes the input for the simulation module.

PHASE 5 (AGENT-BASED MODEL (ABM MODULE))

The Agent-Based Model simulates real hospital behavior by representing the system as interacting independent agents. Patient agents are generated based on the congestion level predicted by the ANN and represent individuals entering the emergency room system. Doctor agents represent medical staff who process patients based on availability and priority rules. The interaction between these agents creates a dynamic queue system where waiting time, service rate, and queue length continuously change depending on system load. This module replicates real-world hospital behavior under different congestion conditions.

PHASE 6 (SYSTEM INTEGRATION MODULE)

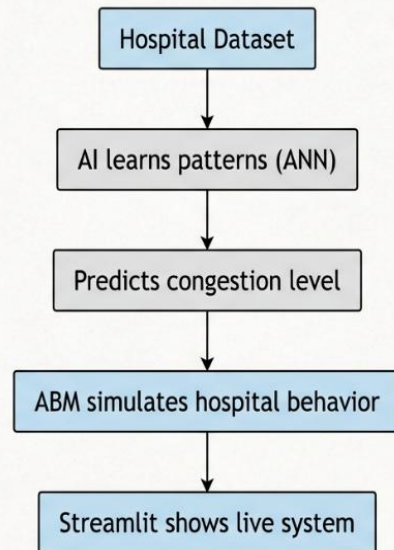
The integration of ANN and ABM forms the core of the system. The ANN acts as the predictive engine that determines the congestion level of the hospital, while the ABM uses this output to control simulation behavior. When the ANN predicts high congestion, the ABM increases queue pressure and patient waiting time. When low congestion is predicted, the system simulates smoother patient flow. This integration ensures that prediction and simulation work together in a unified and realistic system.

PHASE 7 (VISUALIZATION MODULE)

The final stage of the system involves visualization of results through an interactive interface. The system displays congestion levels predicted by the ANN along with simulation outputs generated by the ABM. Users can observe queue behavior, patient flow trends, and hospital workload patterns in a simplified visual format, making the system easy to understand and interpret.

SYSTEM ARCHITECTURE

Hospital Queue Management System Workflow using ANN and ABM



7. ARTIFICIAL NEURAL NETWORK (ANN) MODULE

The Artificial Neural Network in this system is implemented as the core learning and prediction module responsible for analyzing emergency room operational data and transforming it into congestion level classification. This module is trained on structured hospital datasets where each record represents a snapshot of hospital workload conditions defined by variables such as number of waiting patients, number of examined patients, priority distribution, and time-based features extracted from timestamp values. The ANN is designed as a multi-layer feedforward neural network where input features are passed into an input layer and then processed through multiple hidden layers that perform weighted computations to extract hidden patterns from the data. Each hidden layer applies activation functions to introduce non-linearity, enabling the model to learn complex relationships between hospital workload and congestion behavior that cannot be captured using simple rule-based systems. During training, the model adjusts its internal weights using backpropagation to minimize prediction error between actual and predicted congestion states, allowing it to gradually improve accuracy and generalization. Once trained, the ANN acts as a decision-making engine that receives real-time hospital input data and produces an output classification in the form of congestion levels such as low, medium, and high. This output is not an end result but is directly used as an input control signal for the simulation module, making the ANN a predictive intelligence component within the overall system architecture.

8. AGENT-BASED MODEL (ABM) MODULE

The Agent-Based Model in this system is implemented as the simulation engine that replicates real-world emergency room behavior by modeling the hospital environment as a collection of autonomous interacting agents. This module defines two primary agent types, patient agents and doctor agents, where patient agents represent individuals entering the emergency room with varying urgency levels based on predefined priority categories and doctor agents represent medical staff responsible for treating patients according to availability and priority rules. Unlike traditional queue systems, ABM does not rely on fixed equations but instead simulates system behavior through interactions between these independent agents over time. The module receives congestion level output from the Artificial Neural Network, which determines the intensity of patient arrival rates and overall system load. Based on this input, patient agents are generated dynamically and enter the queue system where they are organized according to priority rules, ensuring that critical patients are served first. Doctor agents continuously interact with the queue by selecting patients for treatment and updating system states such as waiting time, queue length, and service completion rates. As these interactions continue, the system evolves in real time, producing a dynamic simulation of hospital operations under different congestion conditions. This module effectively transforms static predictions from the ANN into a realistic behavioral simulation, allowing the system to represent how emergency rooms actually function under varying workloads.

9. SYSTEM INTEGRATION (ANN + ABM WORKING TOGETHER)

The integration of Artificial Neural Network and Agent-Based Modeling forms the core intelligence structure of this system where both prediction and simulation are combined into a single functional pipeline. The ANN module first processes hospital data and produces a congestion classification output that represents the current operational state of the emergency room. This output is not treated as a standalone result but is directly passed into the ABM module as a controlling parameter for simulation behavior.

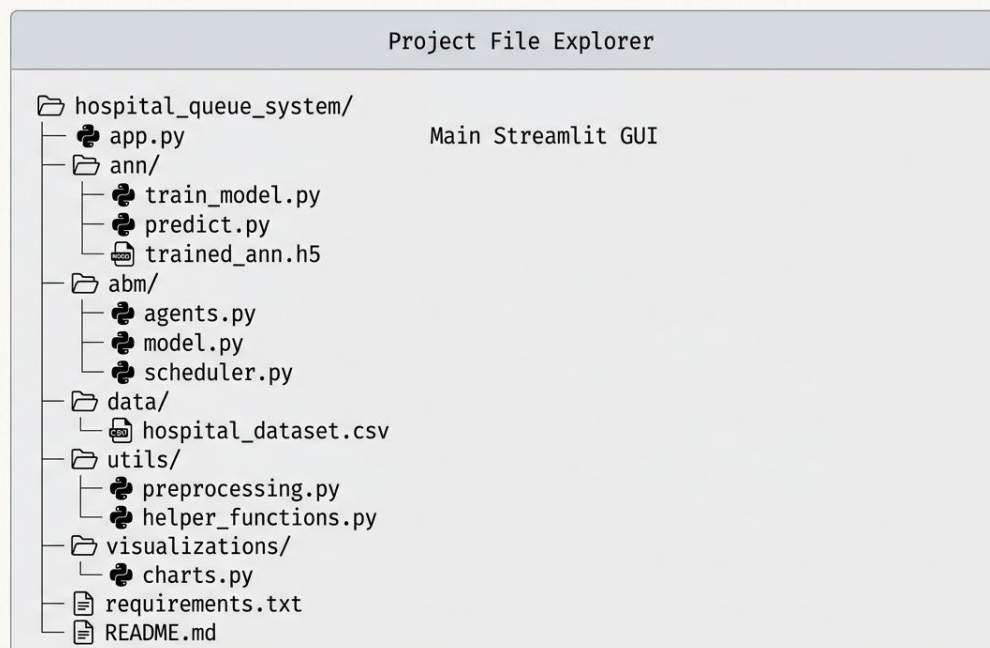
Once the congestion level is generated, it determines how the hospital environment should behave under that specific condition. If the ANN predicts low congestion, the ABM simulates a stable environment where patient arrival rate is low, queues remain short, and doctor agents can efficiently manage patient flow. In the case of medium congestion, the system simulates moderate pressure where queue length increases and waiting time starts to vary depending on patient priority. When high congestion is predicted, the ABM reflects a stressed hospital environment where patient arrival rate increases significantly, queues become longer, and system load intensifies, affecting overall service time.

This integration ensures that the system does not operate in isolation but behaves like a real adaptive hospital environment where prediction directly influences simulation. ANN provides intelligence in the form of decision-making, while ABM translates that intelligence into realistic system behavior, making the overall system dynamic, responsive, and closely aligned with real-world emergency room operations.

10. FOLDER STRUCTURE

The folder structure of the hospital queue management system is designed in a modular way so that each component of the system remains separated, maintainable, and easy to develop independently. The main project directory contains the core application file which is responsible for running the Streamlit interface and connecting all system modules together. Separate modules are created for the Artificial Neural Network and Agent-Based Model so that prediction logic and simulation logic remain independent but can interact through a controlled pipeline. A dedicated dataset folder is used to store raw hospital data including emergency room and attendance records, which are later processed during execution. The preprocessing component is organized in a utility module that handles data cleaning, encoding, and feature engineering tasks before passing data to the ANN model. The trained machine learning model is stored separately in a models directory so that it can be reused without retraining. Another directory is reserved for storing generated outputs such as charts and visualizations, which are used in the dashboard to represent system behavior in a graphical format. This structured organization ensures clarity in development, simplifies debugging, and allows future enhancements without affecting the entire system.

Folder Structure Representation



11. TOOLS AND LIBRARIES

- Python: Core programming language used for system development
- Pandas: Data manipulation and preprocessing of dataset
- NumPy: Numerical computations and array processing
- TensorFlow: Deep learning framework for ANN model development

- Keras: High-level API used for building and training neural networks
- Mesa: Framework used for Agent-Based Modeling (ABM) simulation
- Streamlit: Web-based framework for interactive dashboard (GUI)
- Matplotlib: Data visualization and graph generation
- Visual Studio Code: Development environment used for implementation

12. GUI DESIGN

The system uses a Streamlit-based interface that allows users to interact with the hospital queue management system in real time. The user can select hospital data, view ANN-based congestion predictions, and observe ABM-based queue simulation results. The interface displays outputs in the form of charts, queue status, and congestion levels, making the system easy to understand and use.

13. EXPECTED OUTCOMES

The system is expected to successfully predict hospital congestion levels and simulate real-world emergency room queue behavior. It will provide insights into patient waiting times, queue formation, and hospital workload patterns. The final output will help visualize how emergency rooms behave under different load conditions.

14. CONCLUSION

This project integrates Artificial Neural Networks and Agent-Based Modeling to build an intelligent hospital queue management system. ANN provides prediction of hospital congestion, while ABM simulates real-time patient flow. Together, they create a dynamic system that improves understanding of hospital operations and queue behavior.

15. REFERENCES

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TensorFlow Documentation – <https://www.tensorflow.org>
Mesa Framework Documentation – <https://mesa.readthedocs.io>
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Pandas Documentation – <https://pandas.pydata.org>